String Handling functions in C

String Handling Functions in C

Few commonly used string handling functions are discussed below:

Function	Work of Function
strlen()	computes string's length
strcpy()	copies a string to another
strcat()	concatenates(joins) two strings
strcmp()	compares two strings
strlwr()	converts string to lowercase
strupr()	converts string to uppercase

C – String Handling Functions

strlen()

- To get the length of a string, you can use the strlen() function:

Syntax:

strlen(string_name)

Eg:-

```
char alphabet[] = "ABCDEFGHIJKLMNOPQRSTUVWXYZ";
printf("%d", strlen(alphabet));   // 26
printf("%d", sizeof(alphabet));   // 27
```

We can use `sizeof` to get the size of a string/array.
But `sizeof` and `strlen` behaves differently, as `sizeof` also includes the `\0` character when counting:

C – String Handling Functions

strcat()

- To concatenate (combine) two strings, you can use the strcat() function:

Syntax:

strcat(first_string, second_string)

Eg:-

```
char str1[20] = "Hello ";
char str2[] = "World!";

// Concatenate str2 to str1 (result is stored in str1)
strcat(str1, str2);

// Print str1
printf("%s", str1);  //output: Hello World
```

C – String Handling Functions

strcpy()

- To copy the value of one string to another, you can use the strcpy() function:

Syntax:

strcpy(destination, source)

Eg:-

```
char str1[20] = "Hello World!";
char str2[20];

// Copy str1 to str2
strcpy(str2, str1);

// Print str2
printf("%s", str2);  // Output – Hello World
```

C – String Handling Functions

strcmp()

- To compare two strings, you can use the strcmp() function.
- It returns 0 if the two strings are equal, otherwise a value that is not 0:

Syntax:

strcmp(first_string, second_string)

Eg:-

If str1 is less than str2, the return value is less than 0.
If str1 is greater than str2, the return value is greater than 0.
If str1 is equal to str2, the return value is 0.

```
char str1[] = "Hello";
char str2[] = "Hello";
char str3[] = "Hi";

// Compare str1 and str2, and print the result
printf("%d\n", strcmp(str1, str2));  // Returns 0 (the strings are equal)

// Compare str1 and str3, and print the result
printf("%d\n", strcmp(str1, str3));  // Returns a non-zero value (the strings are not equal)
```